



HFDB-4201 SC

Super-Clean Crosslinkable Power Cable Insulation Compound

HFDB-4201 Super-Clean (SC) is a low-density, crosslinkable, unfilled polyethylene compound designed for high voltage (69kV and higher) power cable insulation requiring the highest degree of cleanliness. It has an extremely low level of contamination and bears the designation SC (Super-Clean). HFDB-4201 SC has been designed to have an extremely low level of additive bloom for a very long storage life and low dusting along with an enhanced degree of scorch retardance for fine mesh filtering and long production run lengths during cable manufacture.

Specifications

Cables insulated with HFDB-4201 SC, using sound commercial fabrication practice, should meet the latest editions of the following industry cable specifications:

- ICEA: S-66-524, S-94-649, S-93-639
- AS: 1429
- BS: 6622
- AEIC: CS7, CS8
- VDE: 0273
- IEC: 60502 and 60840
- EDF NF C-33-223
- REA: 50-70 (U-I)

Properties	Test Methods ⁽¹⁾	Typical Values ⁽²⁾
Physical		
Density	ASTM D 1505	0.92 g/cm ³
Tensile Strength	ASTM D 638	2800 psi (19 MPa)
Tensile Elongation	ASTM D 638	500%
Degree of Crosslinking, Extractables	ASTM D 2765A	<20%
Hot Creep/Hot Set	ICEA T-28-562	<100% / <10%
Electrical		
Dielectric Constant, 60 Hz	ASTM D 150	2.3
Dissipation Factor, 60 Hz	ASTM D 150	0.0003
DC Volume Resistivity, 23°C	ASTM D 257	>1x10 ¹⁵ ohm-cm
Dielectric Strength, 125 mils (3.175 mm)	ASTM D 149	660 V/mil (26 KV/mm)
Short Time (S/T)		590 V/mil (23 KV/mm)
Step-by-Step (S/S)		

(1) Tests are made in accordance with current ASTM or Dow Methods.

(2) These are nominal property values representing tests on molded stress-relieved slabs. Cure times were 15 minutes at 175°C. Data should not be used for specification work. Values are typical, and not to be construed as specifications.

Super-Clean Requirements

HFDB-4201 SC is required to meet strict standards for cleanliness. Throughout the production process, the product is tested to ensure our highest level of cleanliness. Extruded tapes are scanned by an automatic inspection system in a class 10,000 clean room. The purity data is managed using an acceptance sampling plan which ensures that the shipping container meets or exceeds super-clean standards.

Maximum Allowable in 1.0 Kg Tape

	100-250 microns	>250-625 microns	>625 microns
HFDB-4201 SC	0	0	0

In addition, as HFDB-4201 SC is packaged, a continuous side stream of pellets is analyzed using an electronic pellet inspector. A review of detected contamination is incorporated into our SC quality program.

Processing Techniques

HFDB-4201 SC provides excellent surface finish and outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures in the range of 240 to 280°F (116 to 138°C) are recommended, although higher melt temperatures are possible on certain equipment with due care. Generally, a minimum 100-60-40-20 screen pack is recommended. However, specific recommendations for processing conditions can be determined when the application and type of processing equipment are known.

Handling Considerations

Material Safety Data Sheets (MSDS) for Wire and Cable Materials are available from The Dow Chemical Company to help customers/users further understand the proper handling of the product.

Before using this product, read and understand the current Dow Material Safety Data Sheet for this product. As various additives and processing aids used in the manufacture of wire and cable have their own safe use profile, their possible influence on handling and disposal must be investigated separately.

Disposal

DO NOT DUMP INTO ANY SEWERS, ON THE GROUND, OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial, and local laws and regulations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator.

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As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess, or manage chemicals or plastics, and that manage used drums.

Storage and Handling

This product is supplied in pellet form. This product may be readily conveyed and bulk-fed through equipment designed for conventional pelleted polyethylene resin, provided the equipment is designed to prevent accumulation of the fines and dust particles that are contained in all polyethylene resins. These fines and dust particles can, under certain conditions, pose an explosion hazard. We recommend that the conveying system used be: (1) equipped with filters of adequate size, (2) operated and maintained in such a manner to ensure that no leaks develop, and (3) grounded adequately. The fines and dust particles contained in all polyethylene resins are defined as Class I Dust (lowest explosive level) under National Fire Protection Association (NFPA)-68. Your facilities and procedures should conform to NFPA-654. These fines and dust particles are considered a nuisance dust under OSHA-1910.1000, and personnel exposure must be controlled accordingly. We further recommend good housekeeping be practiced throughout your facility.

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